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Flutter Tests of the Pazy Wing

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The paper presents the flutter and limit-cycle oscillation (LCO) tests of the Pazy wing, a very flexible wing model designed to study aeroelastic phenomena associated with nonlinear geometry and provide data for nonlinear aeroelastic model validation. The Pazy wing model, analyses, and data from earlier aeroelastic tests are provided by Avin et al. ("Experimental Aeroelastic Benchmark of a Very Flexible Wing," *AIAA Journal*, 2022, pp. 1–24). In the current study, three velocity sweeps were performed at different angles of attack (AoA), during which the wing entered and exited limit-cycle oscillations (LCO). A motion-recovery camera system and fiber-optics Bragg grating sensors were used to track the deformations and strains over the wingspan. The wing encountered LCO when the wingtip deformation was approximately 25% of the span. The onset velocity, frequency, and mode shape varied, depending on the test's AoA. The paper first focuses on the onset of the oscillations, where the onset velocity and oscillation mode-shapes are analyzed. The paper then continues to investigate the LCO characteristics and the nonlinear behavior of the wing. The wing models and experimental data are publicly available through the Third Aeroelastic Prediction Workshop.[**]**

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I. Introduction

IN RECENT years, flight vehicles have trended toward lighter and more flexible configurations. Under flight loads, such flexible configurations might experience large geometrical deformations that cannot be accurately analyzed with the industry-standard linear structural and aeroelastic models. In recent years, there has been a significant effort in the aeroelasticity community to develop and implement novel nonlinear models that can predict the nonlinear aeroelastic characteristics of flexible configurations in a broad range of flight conditions (see a review in [1]). A major area of research is in the development of nonlinear structural models that can accurately predict the response of very flexible configurations that undergo large deformations. Examples of recent efforts are presented in Refs. [2–7].

The large deflection working group (LDWG) within the Third Aeroelastic Prediction Workshop is a recent collaborative effort to study the modeling and testing issues related to very flexible wings. The test case of these efforts is the Pazy wing, a very flexible wing model developed to study aeroelastic phenomena associated with large deformations. The Pazy wing was designed, fabricated, and tested in the Techion [8] to serve as a benchmark and provide experimental data for code validation. The Pazy wing model and data from earlier aeroelastic tests are provided in Ref. [8]. These tests were conducted with a stabilizing mass to avoid flutter and focused mainly on the static aeroelastic response. Several velocity sweeps were conducted at different angles of attack (AoA), at which the wing reached deformation of over 50% span. Computational analyses of the wing were carried out by the Technion, Imperial College, University of Michigan, German Aerospace Center, and NASA Langley, using different modeling tools [9–13]. Subsequent tests were conducted in Delft University of Technology, and Zurich University of Applied Sciences [14,15].

The current paper presents findings from a flutter test campaign of the Pazy wing. The wing was tested at several angles of attack (AoA), at which the airspeed was increased or decreased until the wing entered and then exited limit cycle oscillations (LCO). The flutter onset and offset velocity, frequency, and mode shape varied, depending on the test's AoA. In all cases, the wing encountered LCO at conditions at which the tip deformation was approximately 25% of the span. The paper first focuses on the onset of the oscillations, analyzing the onset velocity and oscillation mode-shapes, and then investigates the nonlinear behavior of the wing during LCO.

II. Wing Description

The Pazy wing has a chord length of 100 mm, a span of 550 mm, and a NACA 0018 airfoil. It was designed to undergo deformations of about 50% span in wind-tunnel tests. The Pazy wing was tested in LCO conditions measuring displacements and strains over the wingspan. Wing deformations were captured by an OptiTrack[††] motion recovery system (MRS) that tracked 24 infrared reflector stickers taped onto the wing, over its leading- and trailing edges and center chord. An additional three reflectors were attached to the wingtip bar. The wing was also instrumented with multiplexed fiber-optic strain-measuring sensors of the Fiber-Bragg-Grating (FBG) type, providing strain measurement at 60 points over the leading- and trailing edges of the spar [8]. The strain measurements were used to evaluate deformations, using a nonlinear strain to displacement algorithm, as described in Ref. [16]. Figure 1 shows the wing with the sensors.

III. Flutter Analysis

Nonlinear flutter analysis was performed with different formulations by the LDWG participants [9–13]. The structural models used include a beam model [10,11,13], a three-dimensional built-up model

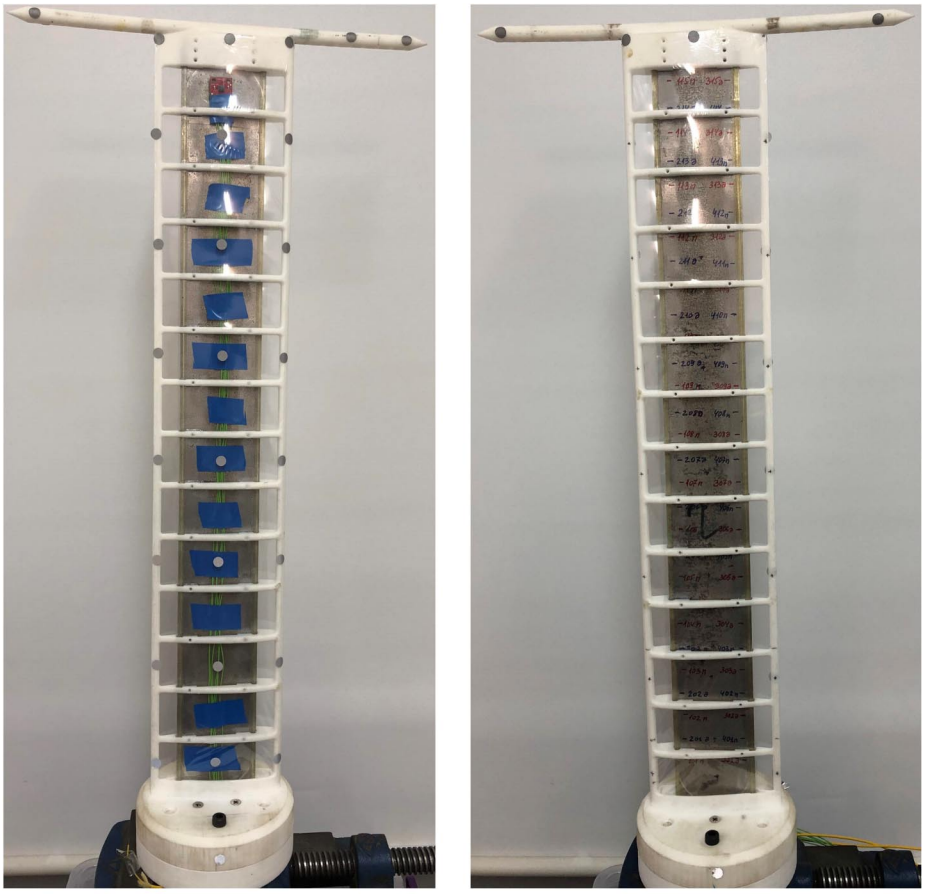
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**Data available online at https://nescacademy.nasa.gov/workshops/AePW3/public.

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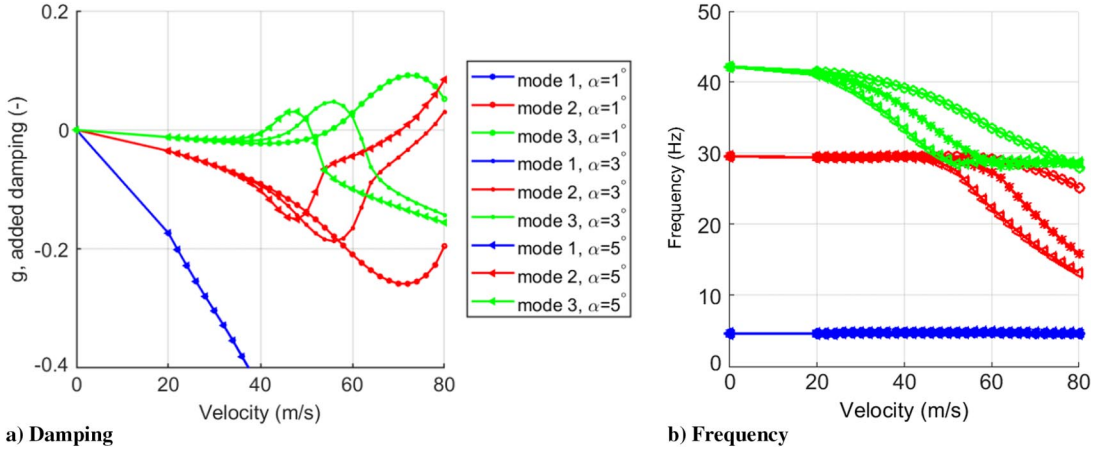


a) b)

Fig. 1 Photo of the Pazy wing and the instrumentation: a) MRS reflectors, and b) FBG strain sensors. Adapted from Avin, O., Raveh, D. E., Drachinsky, A., Ben-Shmuel, Y., and Tur, M., "Experimental Aeroelastic Benchmark of a Very Flexible Wing," ***AIAA Journal*****, 2022, pp. 1–24. doi:10.2514/1.J060621, copyright 2022 Avin, O., Raveh, D. E., Drachinsky, A., Ben-Shmuel, Y., and Tur, M.**

[12], and a modal model [9]. Aerodynamic models of differing fidelity were also used and compared, from a strip-theory model [11], through unsteady potential panel models [9–12], to computational fluid dynamics (CFD) [13].

The current section presents computational results obtained from the modal rotation method (MRM) [9]. The nonlinear MRM relies on a linear modal model from finite-element analysis and the undeformed wing's unsteady aerodynamic modal database. For the Pazy wing study, the modal model was generated from a Nastran finite element method, and the aerodynamic database was obtained from a ZONA6 panel method [17] as a set of generalized, frequency-domain aerodynamic matrices. The MRM first uses a nonlinear modal formulation to compute the static deflections. It then manipulates the generalized matrices of the undeformed structure (both structural and aerodynamic) to calculate the nonlinear dynamics and unsteady aerodynamics of the deformed wing and compute a flutter solution, as detailed in Ref. [9]. The MRM procedure only requires a single calculation of the baseline, undeformed, structural, and aerodynamic models. It is thus highly computationally efficient and capable of computing the whole envelope within seconds, orders of magnitude faster than unsteady vortex lattice method and CFD-based methods, while retaining the main nonlinear features and solution accuracy.



a) Damping b) Frequency

Fig. 2 Matched $V-g$ (left) and $V-\omega$ (right) plots at three AoAs. Adapted with permission from Drachinsky, A., and Raveh, D. E., "Nonlinear Aeroelastic Analysis of Highly Flexible Wings Using the Modal Rotation Method," ***AIAA Journal*****, Vol. 60, No. 5, 2022, pp. 3122–3134. doi:10.2514/1.J061065, Copyright 2020, A. Dravhinsky and D. E. Raveh.**

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Figure 2 shows $\omega - V - g$ plots computed for three AoA values, where each frequency and damping point was computed in a linearized solution about the deformed shape, at the corresponding velocity and AoA. The flutter, which couples the wing's first torsion and second-bending modes, occurs as the frequency of the torsion mode reduces with airspeed to meet the second bending frequency. The damping values in Fig. 2a show that this flutter mode is a hump mode, regaining stability at higher airspeeds, as the torsion frequency reduces to below that of the second bending mode (as seen in Fig. 2b). As the AoA is increased, flutter onset occurs at a lower airspeed, and the hump mode becomes narrower. This is due to the fact that a higher AoA results in a larger wing deformation, increasing the wing's torsional deformation and decreasing its frequency.

IV. Wind-Tunnel Tests

Three velocity sweeps, each at a fixed AoA, were conducted as follows. For test 1, at a root AoA of 7 deg, the airspeed was increased from 0 to approximately 50 m/s to get into and out of the instability region. The AoA was then decreased to 5 deg while keeping the airspeed fixed. The airspeed was decreased to 0, again passing through the unstable region. This exercise was repeated, increasing the airspeed at 5 deg AoA and decreasing it at 3 deg (test 2), and then increasing the airspeed at 3 deg AoA and decreasing it at 1 deg (test 3). This is illustrated in Fig. 3, which shows the loading path atop a stability map computed for the wing at different AoAs. The wind-tunnel airspeed and AoA variation during the tests are shown in Fig. 4a. We note that in several cases there were slight differences between the commanded and actual AoA values. These are evident in small variations in the AoA (about 0.1 deg) and in the overshoots at the AoA reduction.

Due to the large erratic oscillations of the wing, some of the MRS and fiber optic sensor (FOS) measurements were corrupted. The MRS system experienced tracking errors, likely due to reflections from the wing's skin, high velocities, and line-of-sight loss. At low airspeeds and small deformations, these tracking errors were relatively small. The corrupted FOS data is likely due to the large values of oscillatory strains.

Figure 5 shows sample data from test 1, from a single-strain sensor located 127 mm from the root on the main spar, as a function of time, together with the airspeed variation. The strain values in Fig. 5 are multiplied by -1 to make them easier to read with the airspeed data. The wing encountered LCO twice, at 7 deg AoA while increasing the airspeed and at 5 deg while decreasing the airspeed. The AoA was reduced from 7 to 5 deg at approximately 85 s, as evident in the sharp strain reduction. The quasi-steady variation of the strain corresponds to the increase in the dynamic pressure (airspeed increase) with time. The intense strain oscillations correspond to periods of LCO wing motion. We note that the oscillation amplitudes at 5 deg (second LCO) are larger than at 7 deg (first LCO).

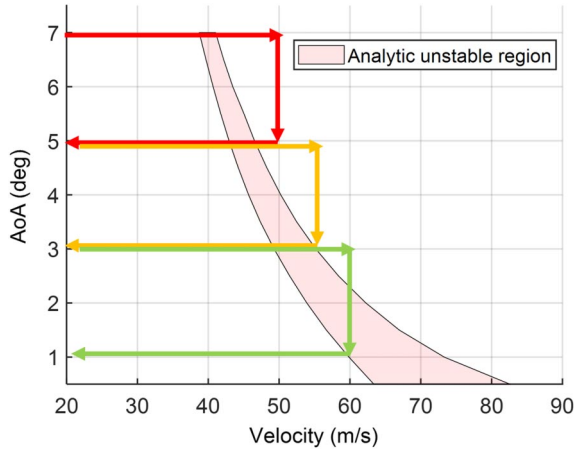


Fig. 3 Analytic stability map and schematic illustration of the test loading path.

Figure 6 shows the strains vs airspeed at 5 deg while increasing and decreasing the airspeed (data from test 2 and test 1, respectively). Hysteresis is observed in the LCO onset and offset conditions. This indicates a subcritical bifurcation in which multiple solutions can be obtained at a certain velocity, depending on the initial conditions.

A. Flutter Characteristics

Figure 7 shows the wingtip displacement and twist-angle time histories for test 1. Because the MRS data was partially corrupted, the figure also shows an estimate of the displacements computed from the measured strain data using a nonlinear strain to displacement transformation algorithm [16]. In Fig. 7a, the flutter onset is marked in a rectangle and the insert plot shows the growth in oscillation amplitude as captured by the MRS and estimated from strains. The MRS data was too erratic to estimate the wingtip twist angle adequately. Thus, for further analyses, we used mostly the strain-based data, whereas the MRS data mainly served for calibration and as a validity check during the preflutter initial phase of each test.

Test 3 differs from the other two tests. In this test, the second flutter was encountered while reducing the AoA from 3 to 1 deg, at fixed airspeed (and not in a fixed AoA during velocity reduction as in the other two tests). The AoA at which flutter started was assessed from the wingtip displacements and AoA. As the AoA was decreased rapidly (over about 1 s), the AoA at which oscillations started could not be identified exactly and is estimated to be between 2 and 2.5 deg.

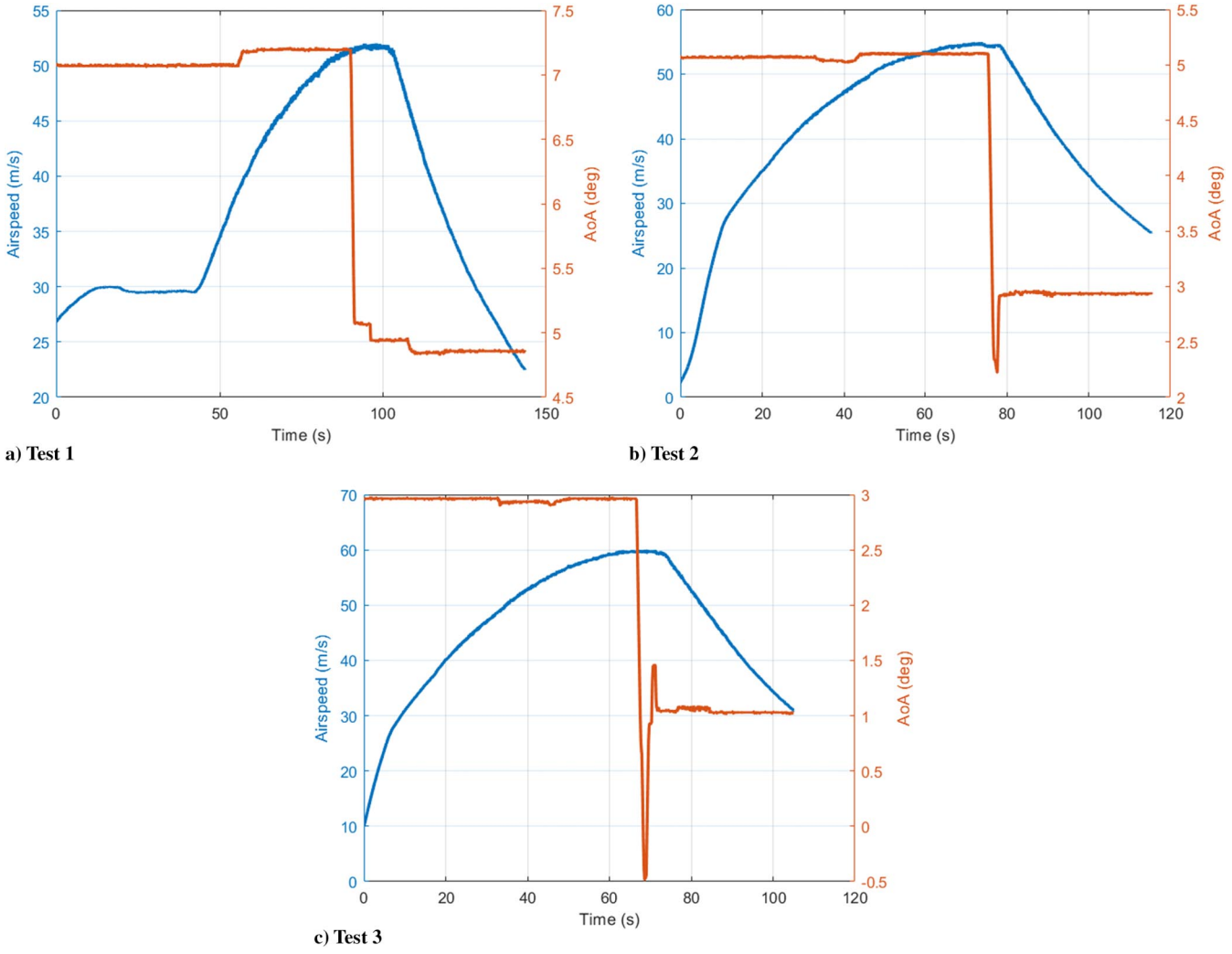
The LCO onset and offset were determined from the wingtip displacements and twist angles' time histories. They are summarized in Table 1 and compared with the analytic prediction by the MRM [9]. The arrows next to the AoA values indicate whether the test was of increasing or decreasing airspeed. Figure 8 shows the flutter-onset speeds from the test, plotted on top of a stability map from the MRM analysis. The red and green triangles indicate flutter onset and offset, respectively. The legend indicates whether the onset/offset occurred while increasing ("Test onset up") or decreasing ("Test onset down") the airspeed. Table 1 and Fig. 8 show a good agreement between the computed and test-onset speeds. Still, the analysis cannot predict the offset speed, as it is based on linearization about a steady condition, which is not the case for the offset. Predicting the offset behavior is significantly more complicated and may require time-domain, harmonic-balance, or asymptotic analyses, and would probably require modeling that includes dynamic stall effects.

Another interesting aspect to examine is the frequency and mode shape at the start of the oscillations, when the nonlinearity of the response is relatively weak, which can be approximately considered as the flutter frequency and mode shape. Figure 9 shows the oscillatory wingtip displacement and twist angle close to flutter onset at 5 deg AoA during up-and- down sweeps. The response data were high-pass filtered at 15 Hz to capture only the oscillatory part and filter out the quasi-steady and first-mode responses. From these plots, the ratio between the wingtip deflection and twist angle can be derived, as well as the phase angle between them. These are summarized in Table 2. On the upsweep, the wingtip twist lags the displacement by 113 deg, whereas on the downsweep the lag is 132 deg.[‡‡] We note that this phase is opposite to a typical first-bending first-torsion flutter mechanism, in which the twist angle leads the displacement. This is because the current mechanism involves second-bending motion, in which the displacement over most of the wing is out of phase with the wingtip motion. Hence, the phase between the modes (denoted "modal phase" in Table 2) can be defined by subtracting 180 deg from the phase between the tip displacement and twist. Based on data from several points on the wing, the accuracy of the data in Table 2 is 0.25(deg/%span) and 5 deg in the amplitude ratio and phase angle, respectively.

From Table 2, it is seen that for the same AoA, all the flutter characteristics are different between the up-and-down velocity sweeps. At the higher speeds the frequency is lower, and the twist-to-displacement ratio and modal phase angles are significantly smaller. Small deviations are also seen between flutter

[‡‡]Positive phase is defined when the twist lags the displacement.

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a) Test 1 b) Test 2

c) Test 3

Fig. 4 Airspeed and AoA variations at the flutter tests.

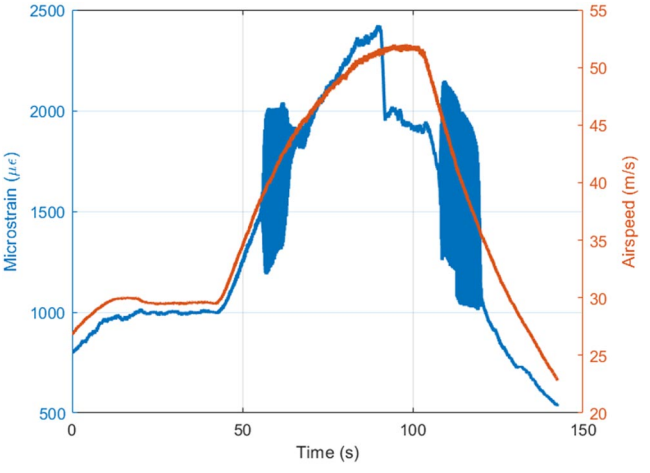


Fig. 5 Strain at a single sensor and airspeed variation during test 1.

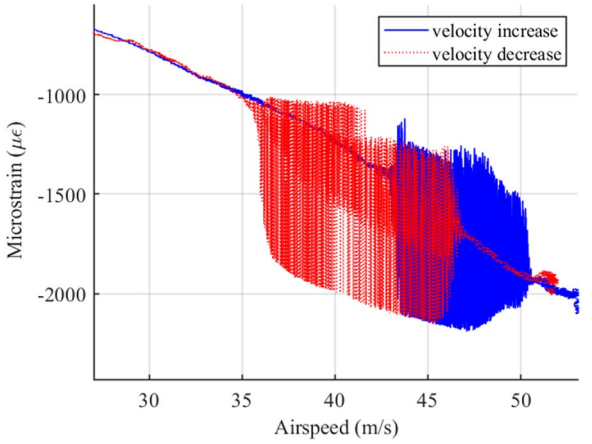


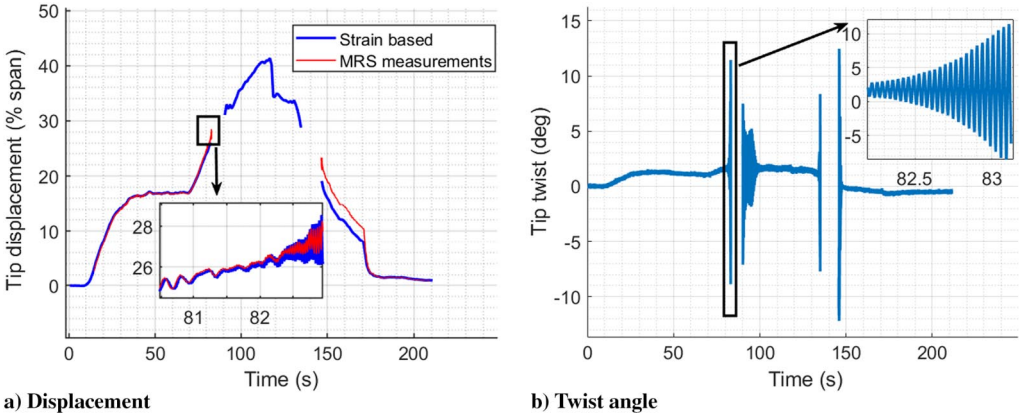
Fig. 6 Strain vs airspeed at LCO 5 deg AoA; LCO entered while increasing/decreasing the airspeed.

modes at different AoAs. The flutter frequency slightly increases with the AoA (and deformation; see Table 1), as does the twist-to-displacement ratio, whereas in the phase angle, no clear trend can be identified.

The displacements and twist angles measured in flutter suggest that the primary source of nonlinearity in the system, which governs the LCO, is aerodynamic. Although the tip deflection dynamic amplitude in the measured region is only a few percent of the span, the wingtip twist angles oscillate at an amplitude of approximately 10 deg, at which the airfoil is expected to stall.

B. LCO Characteristics

The measured strain data from an arbitrary sensor was analyzed for its frequency content. Figure 10 shows spectrogram plots of the data at the three tests, presenting the variation of the frequency content of the signal with airspeed. The LCO regions are identified from the



a) Displacement b) Twist angle

Fig. 7 a) Wingtip displacement, and b) twist angle at test 1.

Table 1 Flutter onset/offset airspeeds

Test	AoA, deg	Analytic flutter onset velocity, m/s	Test flutter onset velocity, m/s	Error, %	Test flutter offset velocity	Flutter onset deflection, % span
1	7 ↑	38.7	38	1.8	46	26
1	5 ↓	46.5	48	3.1	36	31.5
2	5 ↑	42.9	43	0.2	51	24.5
2	3 ↓	55.2	55	0.4	40	28.5
3	3 ↑	49.2	49	0.4	58	23
3	2 – 2.5 ↓	62.2–58.4	60	0–3.7	44	29

Table 2 Flutter frequency and mode comparison

AoA, deg	Frequency, Hz	Twist to disp. amplitude ratio, deg/% span	Twist to disp. phase, deg	Modal phase, deg
7 ↑	29.4	5.78	106	−74
5 ↑	29.9	5.16	113	−62
3 ↑	30.0	5.17	97	−83
5 ↓	26.7	3.95	152	−36
3 ↓	28.2	3.44	132	−48
2–2.5 ↓	27.2	3.1	130	50

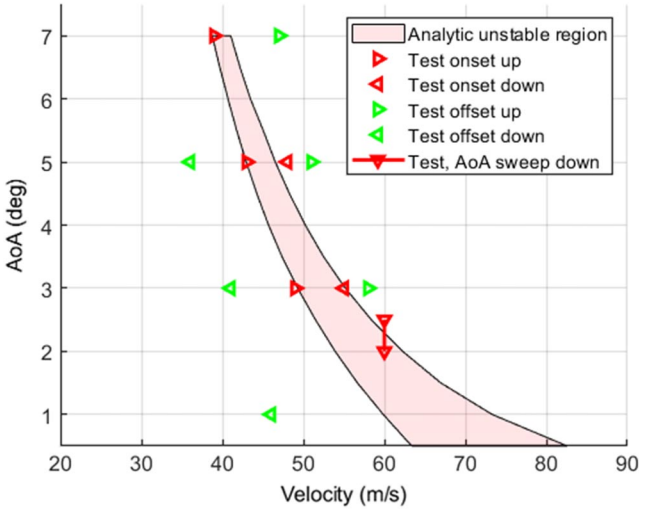
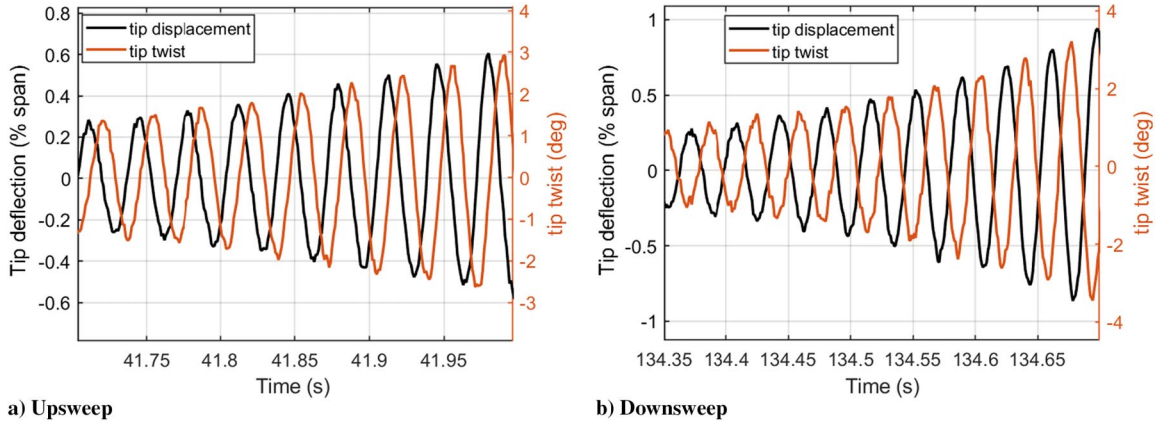


Fig. 8 Comparison of computational and experimental flutter onset/offset airspeeds.

largely amplified first-torsion second-bending modal response, and a strong nonlinearity of the response is evident from the presence of the high harmonics.

To further quantify the participation factor of the different harmonics, the oscillation-time history was divided into subsections and each subsection was fitted with a sinusoidal function of the type $\varepsilon \approx \sum_{i=1}^{3} A_i \sin(\omega_i + \phi_i)$. Sample strain-time histories and their corresponding fitted data are presented in Fig. 11 for test 1 at 7 deg AoA, showing that a three-term multisine function accurately describes the motion. A fourth term is also present in several cases, but its amplitude is small. Figure 12 shows the evolution of the frequencies and amplitudes of the sine terms during the upsweeps. Only the frequency of the first harmonic is shown. The other two are exact multiples of the first (this is a result of the fitting, not a constraint). Results are shown for two arbitrary strain sensors in each of the three tests. The significance of the higher harmonics is evident, as their amplitudes are of the same order of magnitude as the main frequency, and even higher in some cases. Figure 12 shows that the oscillation frequency reduces with airspeed. It can also be seen that the dynamic response changes



a) Upsweep b) Downsweep

Fig. 9 Oscillatory wingtip displacement and twist angle during flutter onsets at AoA of 5 deg.

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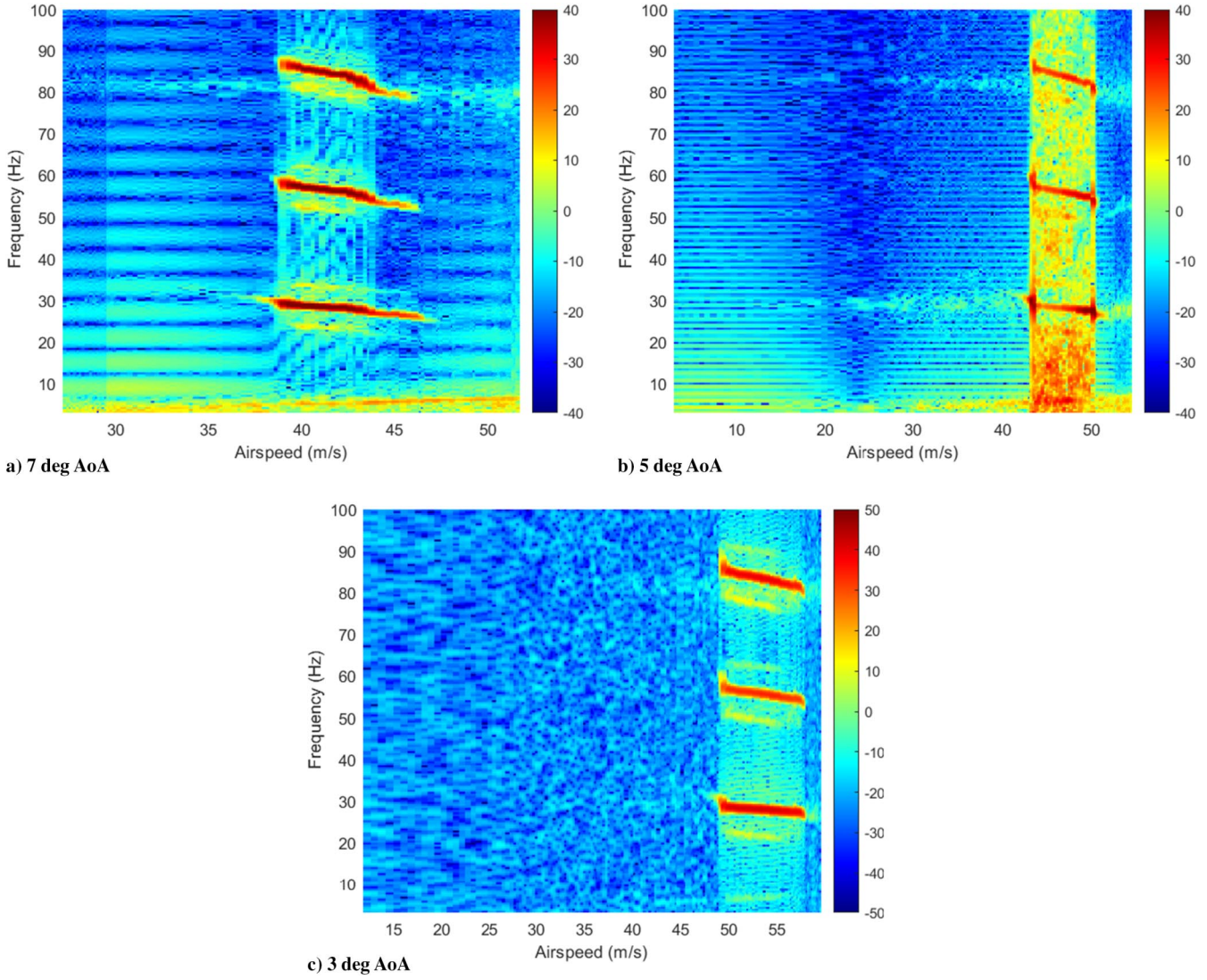


Fig. 10 Spectrograms of strain data from the three tests.

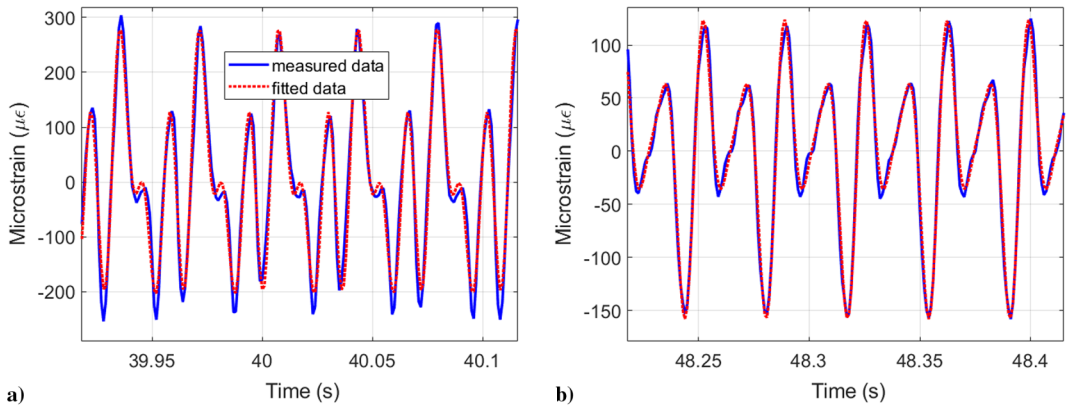


Fig. 11 Comparison of two sample segments of the strain history (at different arbitrary time frames) to a fitted multisine function.

with airspeed, not only in the amplitude but also in the ratio among the different harmonics.

V. Conclusions

The current paper presented data analysis from the flutter and LCO tests of the Pazy wing. The test results were analyzed and compared with nonlinear aeroelastic analyses. Three velocity sweeps were performed at different AoAs, during which the wing entered and exited LCO. A motion-recovery camera system and fiber-optic Bragg grating sensors were used to track the deformations and strains over the wingspan. The wing encountered LCO when the wingtip deformation was approximately 25% of the span, at varying onset speed, frequency, and mode shape, depending on the test's AoA. An analysis of the flutter onsets showed large wingtip twist angles of over 10 deg in the flutter mode compared to deformation of only a few percent of the span, suggesting that the primary source of nonlinearity in the oscillating system is aerodynamic stall. The wing's LCO response was nonlinear, as evidenced by the high harmonics' significant frequency

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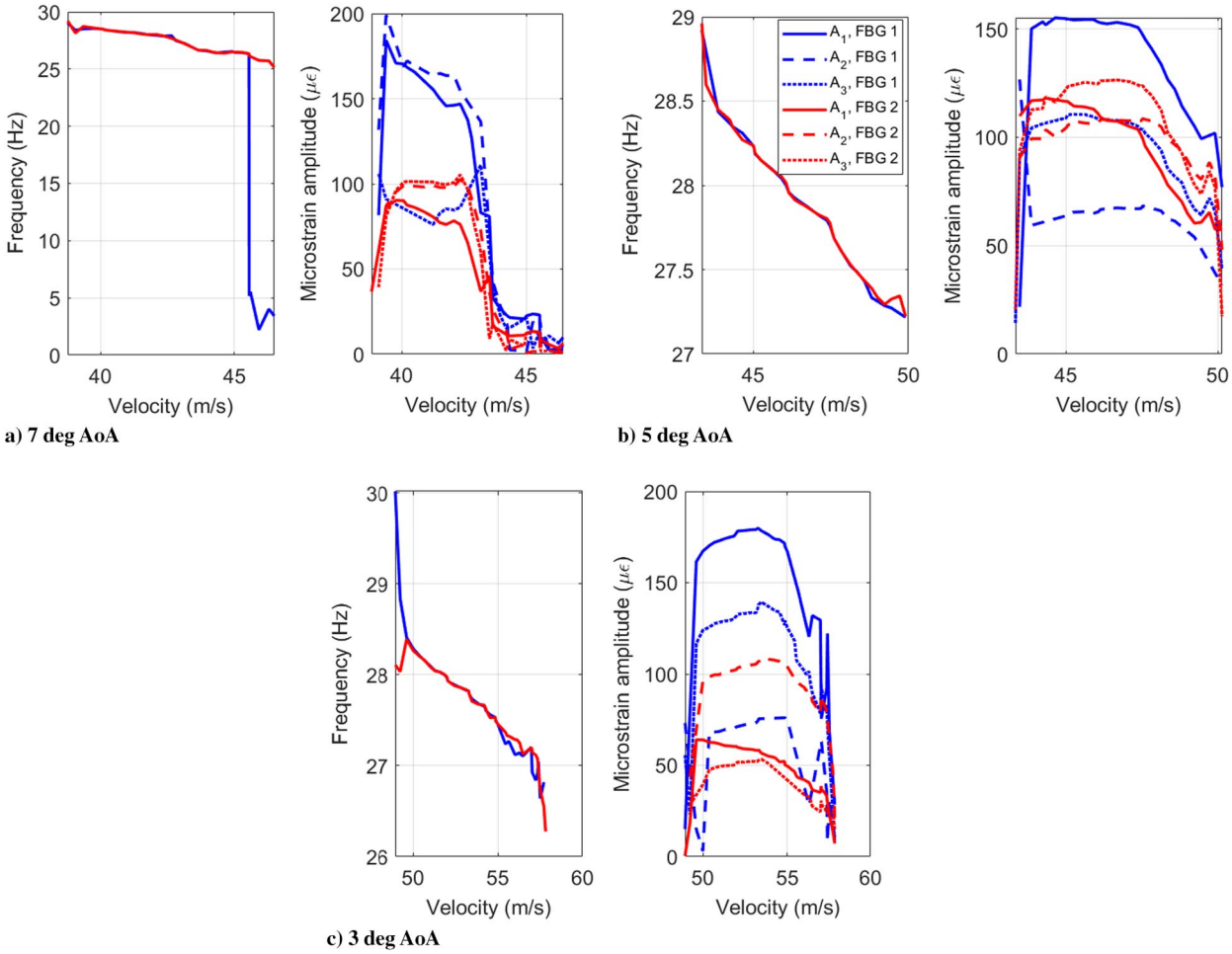


Fig. 12 Frequency of the first harmonic and amplitude of the first three harmonics in the LCO response, as measured by two FBG sensors at the different tests.

content and its evolution with airspeed increase. An increase in AoA led to a decrease in the flutter velocity and a slight reduction in the flutter frequency. Differences were observed between the LCO onset characteristics during velocity upsweep and downsweep. These were mainly in the frequency and in the twist-to-deflection ratio, which were smaller at the downsweeps, and in the smaller phase angle between the displacement and twist modes. Significant hysteresis was observed in the system, leading to differing flutter onset and offset velocities during velocity up-and-down sweeps, attributed to a subcritical type of bifurcation. Strain data analysis showed that during the LCO, the response at a certain airspeed, during upsweep or downsweep, was consistent and independent of the initial conditions despite the hysteretic behavior. All Pazy wing models and the data from this test are publicly available through the Third Aeroelastic Prediction Workshop (see Abstract footnote for URL) to serve in future studies that validate computational models of very large deformations and examine various aspects of the aeroelastic response of very flexible wings.

Acknowledgment

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